

Test Cases

Project: Electronics Store Inventory Management System

Phase 1 — Functional, NFR and Security Test Cases

Item	Details
Version	1.0
Team	H9
Technology Stack	Python (Django)
Status	Draft / For Review

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Part of the Team H9 Phase 1 document set: 01 SRS | 02 Test Plan | 03 Test Cases | 04 Architecture & Design.

Section and requirement numbers follow the master SRS so cross-references (RTM, VFR, DS-xx, TC-xx) remain valid across the four documents.

Test Cases

This document lists the planned test cases referenced by the Requirements Traceability Matrix (RTM) of the master SRS. Each test case is mapped to the requirement(s) it verifies and to the measurable validation criterion defined for that requirement (VFRs for functional requirements; the stated measurement for non-functional requirements; the expected validation of the planned security tests for security requirements).

Status reflects the RTM: no test has been executed in Phase 1 (N = Not tested). Pass/fail evidence will be recorded during implementation and testing. ESIM-NF-006 is verified by code review and has no separate test case.

Test Case ID	Type	Requirement(s)	Expected result / validation criterion	Status
TC-AUTH-01	Functional	ESIM-F-001 — User Authentication	Valid credentials authenticate; invalid credentials do not create an authorized session.	Not tested
TC-AUTH-02	Functional	ESIM-F-002 — Logout & Session Management	After logout, protected inventory operations are denied until re-authentication.	Not tested
TC-INV-01	Functional	ESIM-F-003 — Add Component	A valid component form submission creates exactly one record with the entered values.	Not tested
TC-INV-02	Functional	ESIM-F-004 — View Inventory	Every component in the test dataset is retrievable and displayed with stored details.	Not tested
TC-SEARCH-01	Functional	ESIM-F-005 — Search / Filter	Search/filter results contain only records satisfying the selected criteria.	Not tested
TC-INV-03	Functional	ESIM-F-006 — Update Component	Updating a component changes the selected record without creating a duplicate ID.	Not tested
TC-INV-04	Functional	ESIM-F-007 — Delete Component	Confirmed deletion/deactivation removes the selected component from active inventory.	Not tested
TC-AVAIL-01	Functional	ESIM-F-008 — Availability Status	Availability matches the documented rule for quantities below, equal to and above the threshold.	Not tested
TC-LOW-01	Functional	ESIM-F-009 — Low-Stock Identification	All components at or below minimum stock are identified.	Not tested
TC-DASH-01	Functional	ESIM-F-010 — Dashboard	Dashboard totals equal the corresponding counts calculated from the test dataset.	Not tested
TC-REP-01	Functional	ESIM-F-011 — Inventory Reports	Generated report values match relevant stored component and stock records.	Not tested
TC-VAL-01	Functional	ESIM-F-012 — Input Validation	Missing, duplicate and negative/invalid values are rejected with a clear validation message.	Not tested
TC-DATA-01	Functional	ESIM-F-013 — Persistent Storage	A valid component record remains retrievable after logout and subsequent login.	Not tested
TC-SEC-01	Security	ESIM-F-014 — Authorization ESIM-SR-001 — Security	Protected pages cannot be accessed without authentication. A user without required permission is blocked from restricted inventory changes.	Not tested
TC-ERR-01	Functional	ESIM-F-015 — Error Handling	Failed/invalid operations provide understandable feedback without corrupting valid records.	Not tested
TC-AUD-01	Functional	ESIM-F-016 — Inventory Change Tracking	Where audit tracking is enabled, a supported inventory change creates the defined audit information.	Not tested
TC-PERF-01	Non-functional	ESIM-NF-001 — Performance	For normal academic test load, 90% of inventory list, search and CRUD requests shall complete within 2 seconds.	Not tested
TC-PERF-02	Non-functional	ESIM-NF-002 — Scalability	The system shall support at least 1000 component records in the project test database without functional failure.	Not tested

Test Case ID	Type	Requirement(s)	Expected result / validation criterion	Status
TC-REL-01	Non-functional	ESIM-NF-003 — Reliability	A valid inventory transaction shall either complete successfully or leave the affected record in its previous valid state.	Not tested
TC-REL-02	Non-functional	ESIM-NF-004 — Availability	During a controlled test run, the application shall remain available for at least 99% of the scheduled test duration, excluding planned maintenance.	Not tested
TC-UX-01	Non-functional	ESIM-NF-005 — Usability	A new user shall be able to log in, view inventory and perform permitted common tasks without developer assistance; actions shall provide clear feedback.	Not tested
TC-COMP-01	Non-functional	ESIM-NF-007 — Compatibility	The application shall function correctly on current versions of Chrome, Edge and Firefox.	Not tested
TC-REC-01	Non-functional	ESIM-NF-008 — Recoverability	A database backup of the project inventory shall be restorable in the defined development environment.	Not tested
TC-SEC-02	Security	ESIM-SR-002 — Security	A user without the required role cannot perform restricted inventory operations.	Not tested
TC-SEC-03	Security	ESIM-SR-003 — Security	Passwords and authentication secrets are not stored or logged in plaintext.	Not tested
TC-SEC-04	Security	ESIM-SR-004 — Security	Unauthorized state-changing requests are rejected.	Not tested
TC-SEC-05	Security	ESIM-SR-005 — Security	Invalid or malicious input is rejected without corrupting valid records.	Not tested
TC-SEC-06	Security	ESIM-SR-006 — Security	Application logs shall not record passwords or authentication secrets.	Not tested